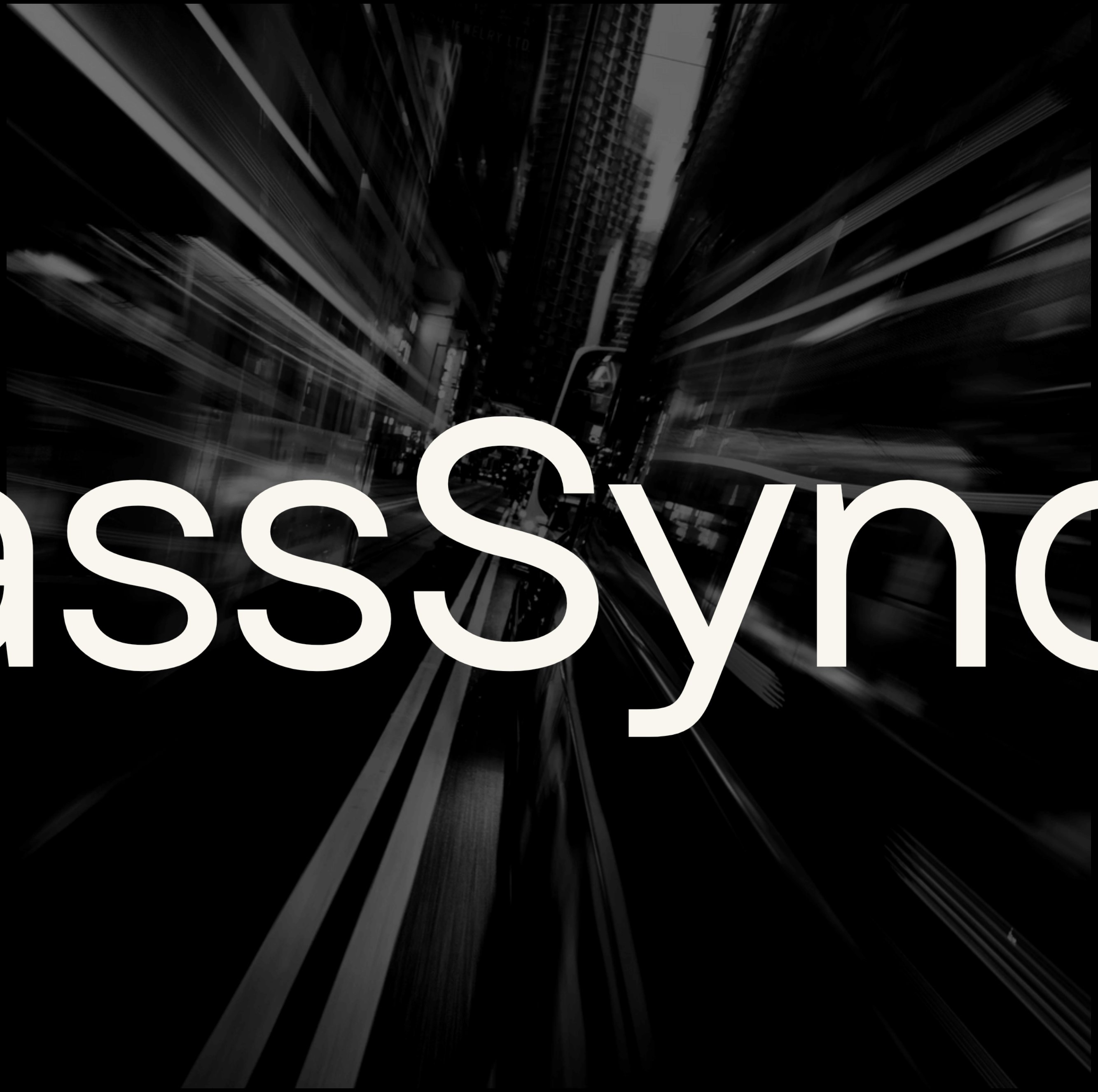
BNU

PROJECT PROPOSAL

SUBMITTED TO: MISS NOREEN KHALID



ClassSync AI

BSCS

v1.0

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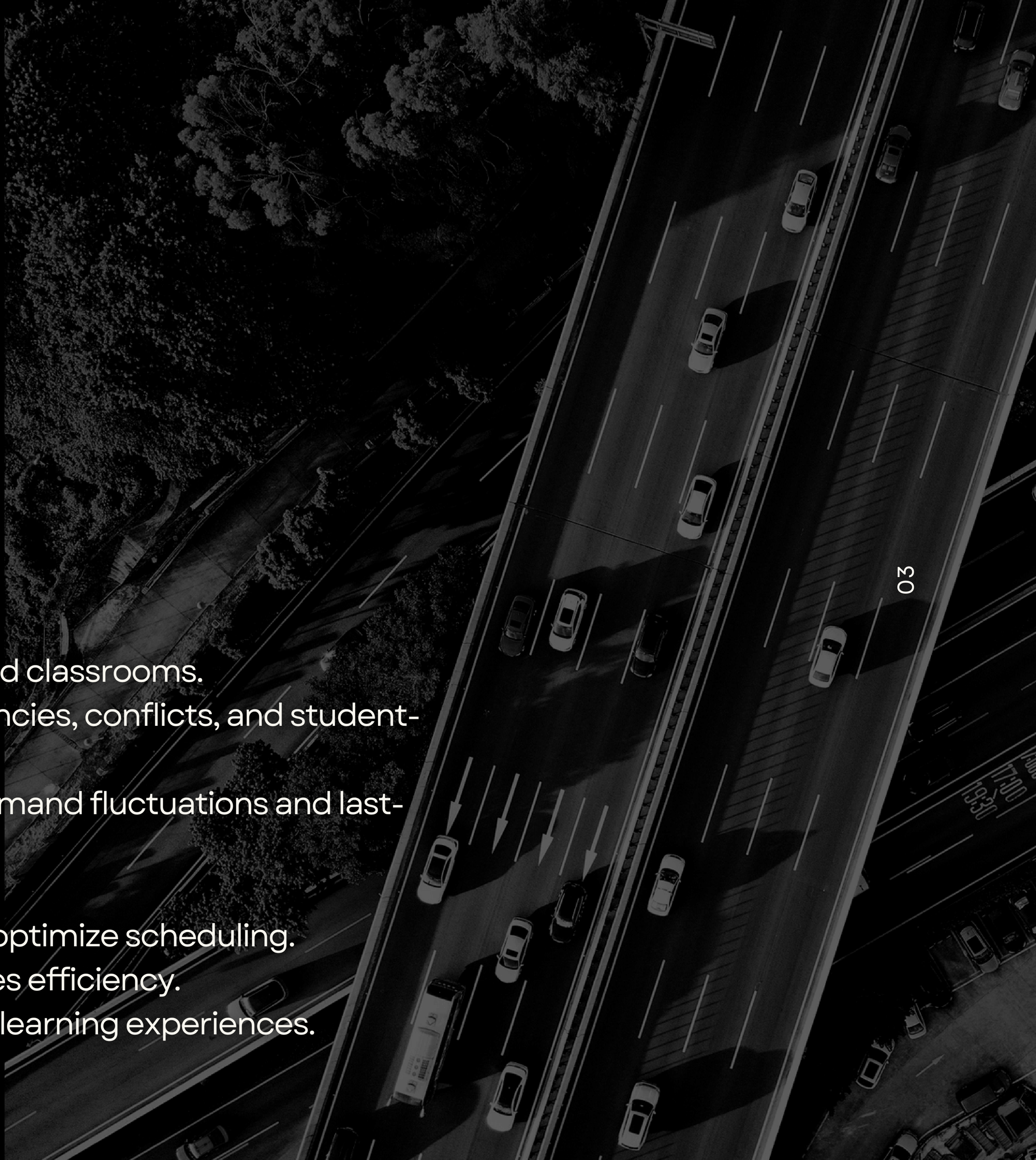
BACKGROUND AND JUSTIFICATION

University course scheduling remains a complex challenge due to overlapping classes, instructor availability, and resource constraints.

- University course scheduling is complex, involving student enrollments, instructor availability, and time slot constraints.
- Traditional scheduling methods are inefficient, leading to conflicts and inconsistencies.
- AI and ML offer a data-driven solution, improving automation, fairness, and scalability.
- An adaptive AI system can optimize timetables, balancing course loads and enhancing resource utilization.

Problem Statement



- University scheduling faces major challenges due to:
  - Overlapping courses, instructor constraints, and limited classrooms.
  - Manual and rule-based scheduling, leading to inefficiencies, conflicts, and student-instructor dissatisfaction.
  - Lack of adaptability to real-time changes, including demand fluctuations and last-minute adjustments.
- This research aims to develop an AI-driven system that:
  - Uses historical data and student enrollment trends to optimize scheduling.
  - Reduces conflicts, balances course loads, and improves efficiency.
  - Enhances faculty workload management and student learning experiences.

Aims and Objectives

- To develop and validate an AI-driven scheduling system that:
  - Optimizes university timetables with minimal conflicts.
  - Enhances scheduling efficiency through automation.
  - Adapts dynamically to real-time constraints.
  - Ensures fair resource allocation for students and faculty.

PREVIOUS RESEARCH GAP

- Existing scheduling methods rely on manual, rule-based, and heuristic approaches, leading to inefficiencies.
- AI-driven scheduling research has focused on industries and workforce management, with limited implementation in universities.
- Existing AI-based solutions lack real-world university testing, making them less adaptable to academic constraints.
- Current methods fail to accommodate real-time changes, fairness in class distribution, and student preferences.
- There is a need for an AI-driven approach that optimizes course allocation, reduces conflicts, and improves scheduling efficiency.

Objectives

- Develop an AI-driven scheduling system that integrates course schedules, faculty availability, student enrollments, and classroom capacities.
- Ensure real-time adaptability by handling course demand shifts, faculty unavailability, and last-minute changes.
- Implement machine learning algorithms for conflict detection, resource optimization, and predictive scheduling.
- Improve student and faculty experience by ensuring conflict-free schedules, fair workload distribution, and balanced class assignments.
- Provide interactive visualization tools for real-time analysis of schedules, course distribution, and resource utilization.
- Enable user feedback integration to refine and optimize the system for better usability and scalability.

QUESTIONS

Research Questions

- How can AI-driven algorithms enhance the accuracy and efficiency of university timetable generation while considering course availability, instructor preferences, and classroom constraints?
- What impact does a data-driven scheduling system have on reducing conflicts, balancing faculty workload, and improving student satisfaction compared to traditional scheduling methods?
- Can machine learning models predict and adapt to changing scheduling constraints, such as faculty availability, student demand, and classroom occupancy, to ensure an optimized and scalable timetable system?

Significance of Research

The AI-driven scheduling system minimizes scheduling conflicts and errors by optimizing faculty allocations, reducing classroom clashes, and improving resource utilization. It enhances student and faculty satisfaction by eliminating long gaps, preventing course overlaps, and ensuring fair workload distribution. As a scalable and adaptable solution, it can adjust to different university structures, faculty preferences, and semester variations. Additionally, it improves academic planning through data-driven insights, making scheduling more efficient and informed. By enabling AI-driven adaptability, the system future-proofs university timetables, allowing institutions to handle last-minute schedule changes without requiring manual rework.

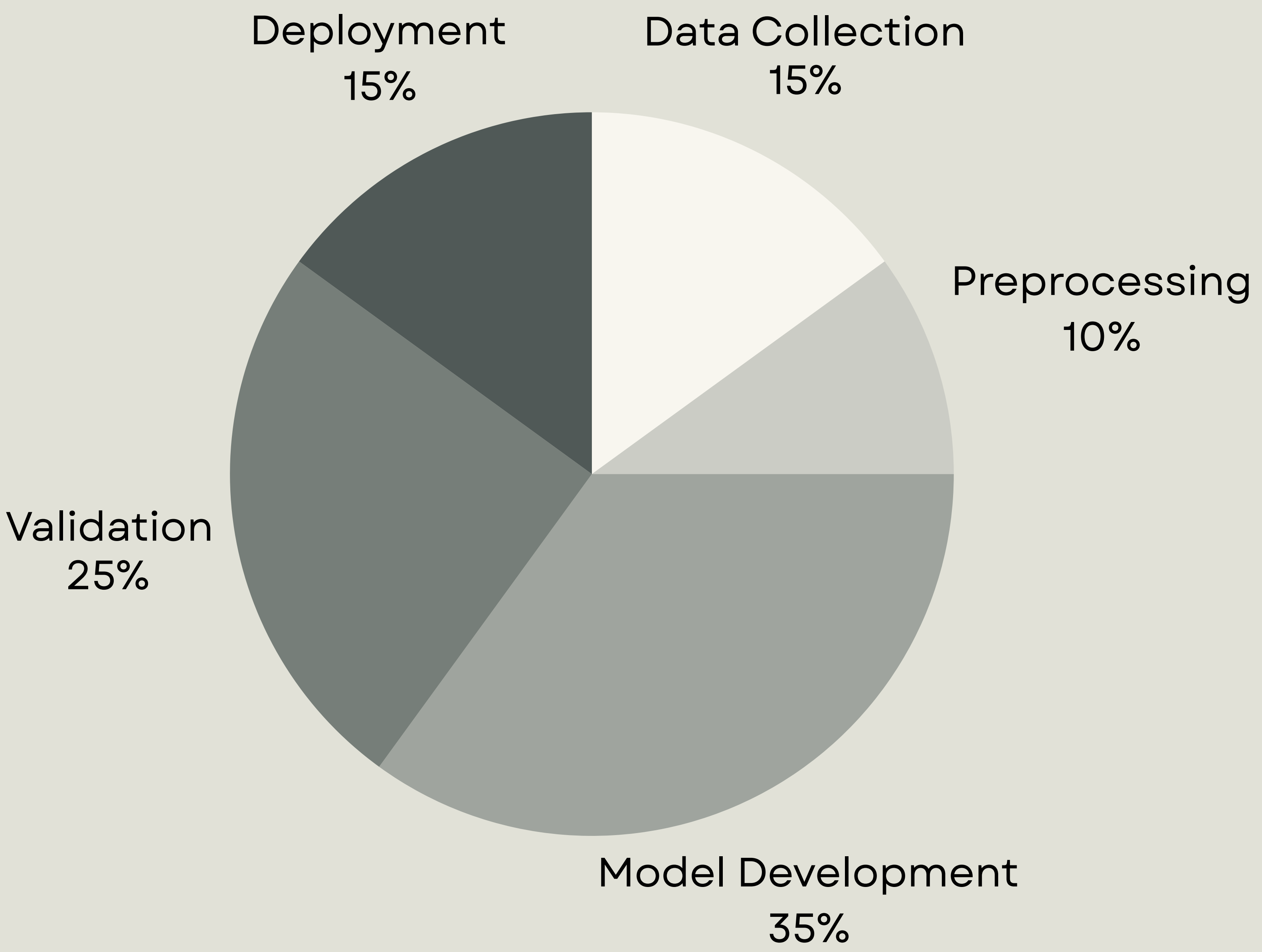
Methodology

To achieve the research objectives, the following steps will be undertaken:

1. **SYSTEM DESIGN AND INTEGRATION**
2. **ALGORITHM DEVELOPMENT**
3. **USER INTERFACE DEVELOPMENT**
4. **SYSTEM IMPLEMENTATION**

Time Spent on Each Project Phase

This pie chart represents the distribution of time allocated to each phase of our project. The largest portion is dedicated to Model Development (35%), as it involves implementing and optimizing AI algorithms. Data Collection & Preprocessing (25%) takes a significant share to ensure clean and structured data. Validation & Performance Testing (20%) is crucial for refining accuracy, while Deployment & Refinement (20%) ensures real-world adaptability. This breakdown highlights the effort required at each stage for successful project completion.



Methodology Plan

The plan outlines our structured approach, starting with Data Collection and Preprocessing to clean and organize scheduling data. Next, Model Development and Validation ensure accurate, AI-driven timetable optimization. The system is then Implemented & Tested in a simulated environment before Deployment for real-world use.

Gathering real-time and historical data

Data Collection

Cleaning, normalizing, and segmenting data

Data Preprocessing

Developing AI models

Model Development

Validating models using cross-validation

Model Validation

Testing the model in a simulated environment

Implementation and Testing

Deploying the final model into a live system

Deployment

Importance

Provides a scalable model for AI-driven academic management.
Enhances efficiency and automation in university scheduling.
Reduces manual workload and scheduling conflicts.
Supports fair and optimal resource allocation.

CONCLUSION

Purpose of Research:

Develop an AI-driven scheduling system to optimize university timetables and minimize conflicts.

Key benefits:

- Automates scheduling for efficiency and fairness.
- Provides real-time adaptability to dynamic constraints.
- Reduces administrative workload and improves resource utilization.

Enhances university scheduling, ensuring a scalable and intelligent approach to academic management.

Thank you